Radar Design Project Proposal

This is an example of a good proposal. Please ignore the values shown as they do not apply to this semester's specs.

Group #

Members (names): XXXXXX

UCF, Spring XXXX

RADAR DESIGN PARAMETERS

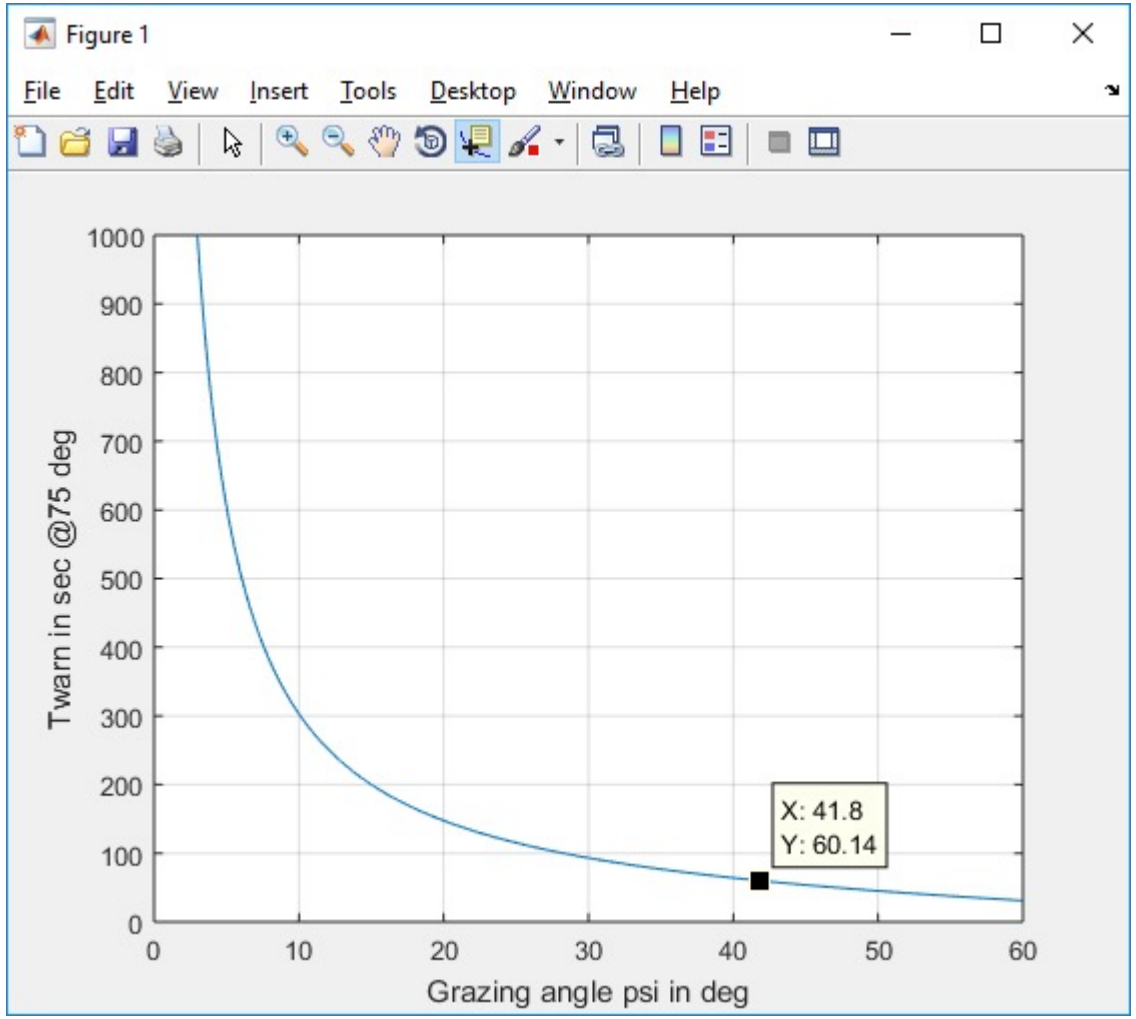
This is the preliminary specification for Spring 2022

Radar Design Parameters Choices		
Radar Freq	XXX	Hz
UAV Altitude	XXX	km
Ant.boresight grazing ang	XXX	deg
Ant dia- elevation	XXX	meter
Ant dia- azimuth	XXX	meter
Ant spin rate	XXX	rpm
Antenna Azimuth @ EOS (measured ± from fore-look)	XXX	deg
Xmtr Pwr	XXX	W
Xmt pulse width	XXX	microsec
PRF	XXX	Hz
<Tfa>	XXX	sec
Design capability: worst case # scans on target = N	XXX	integer
PCR (decimal # ≥ 1.0)	XXX	
Simultaneous Target detection?	X	"Y" or "N"
Doppler-mode PRF	XXX	Hz

Examples of Trade Study Results

The following <u>may or may not apply</u> to this semester's spec

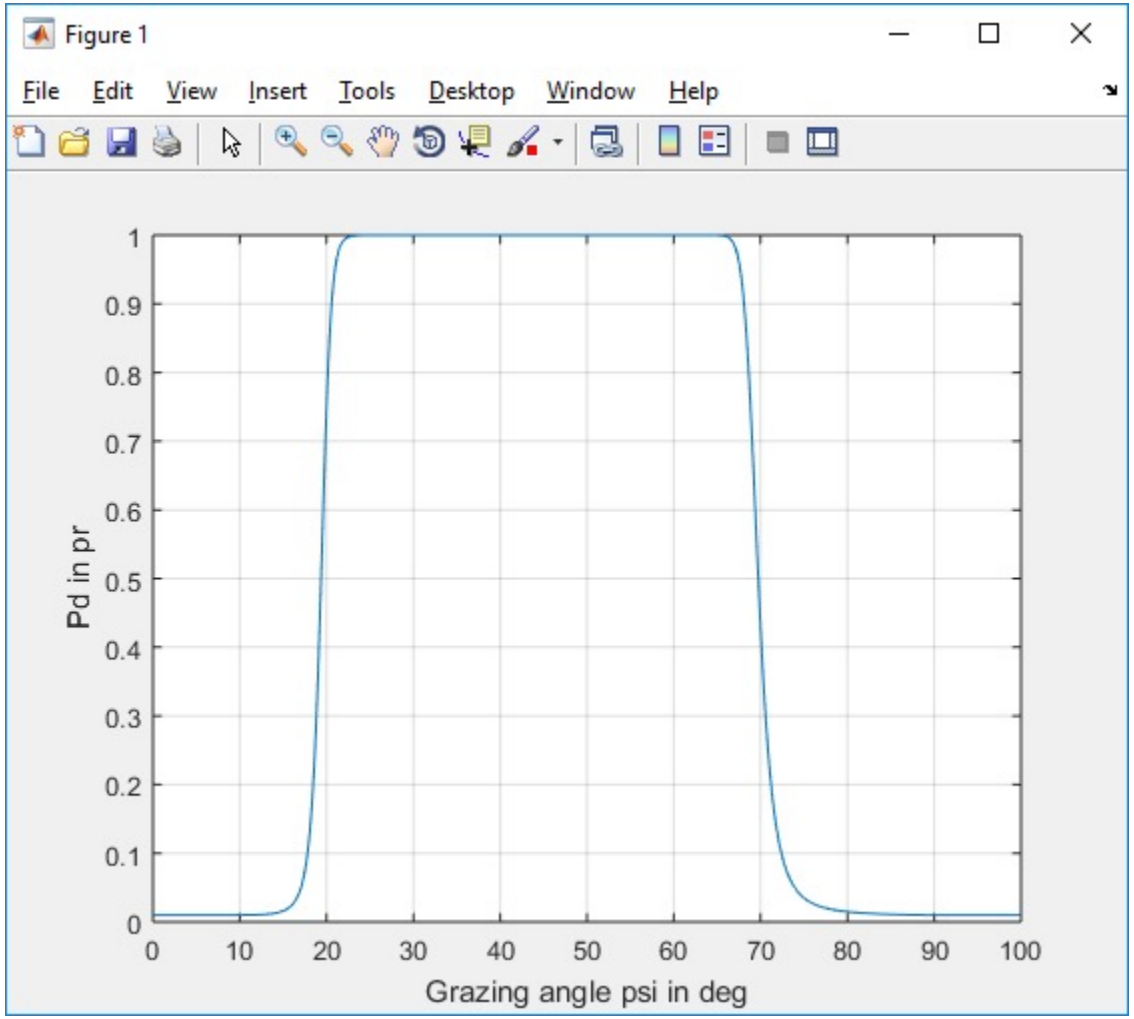
TRADE STUDY Twarn vs Grazing Angle ψ



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see for ψ ≤41.8 deg,
Twarn≥ 60 sec
The lower ψ the longer
Twarn and the lower Pd.
However; due to constrain
for Pd ≥ XX (single scan)
We picked up: ψ= XX deg
Then Twarn= XX sec.
And Pd= XX

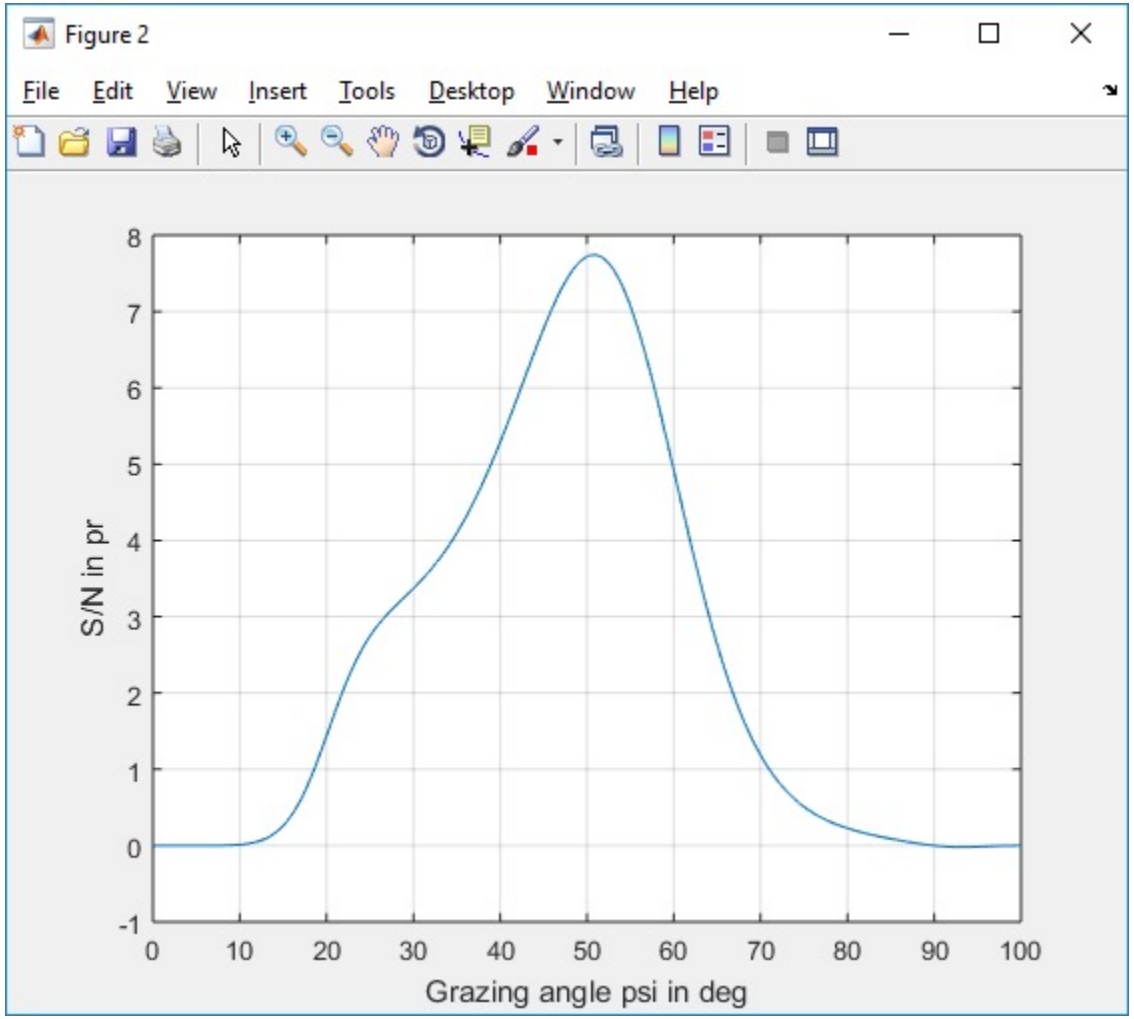
TRADE STUDY Pd vs Grazing Angle ψ



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see for ψ between 19 to 70 deg, Pd ≥ 0.5
The lower ψ the longer Twarn and the lower Pd (for ψ ≤ 30deg).
However; due to constrain for Twarn ≥ 60 sec
We picked up: ψ=20 deg
Then Twarn=147.46 sec.
And Pd=0.7518 p.r.

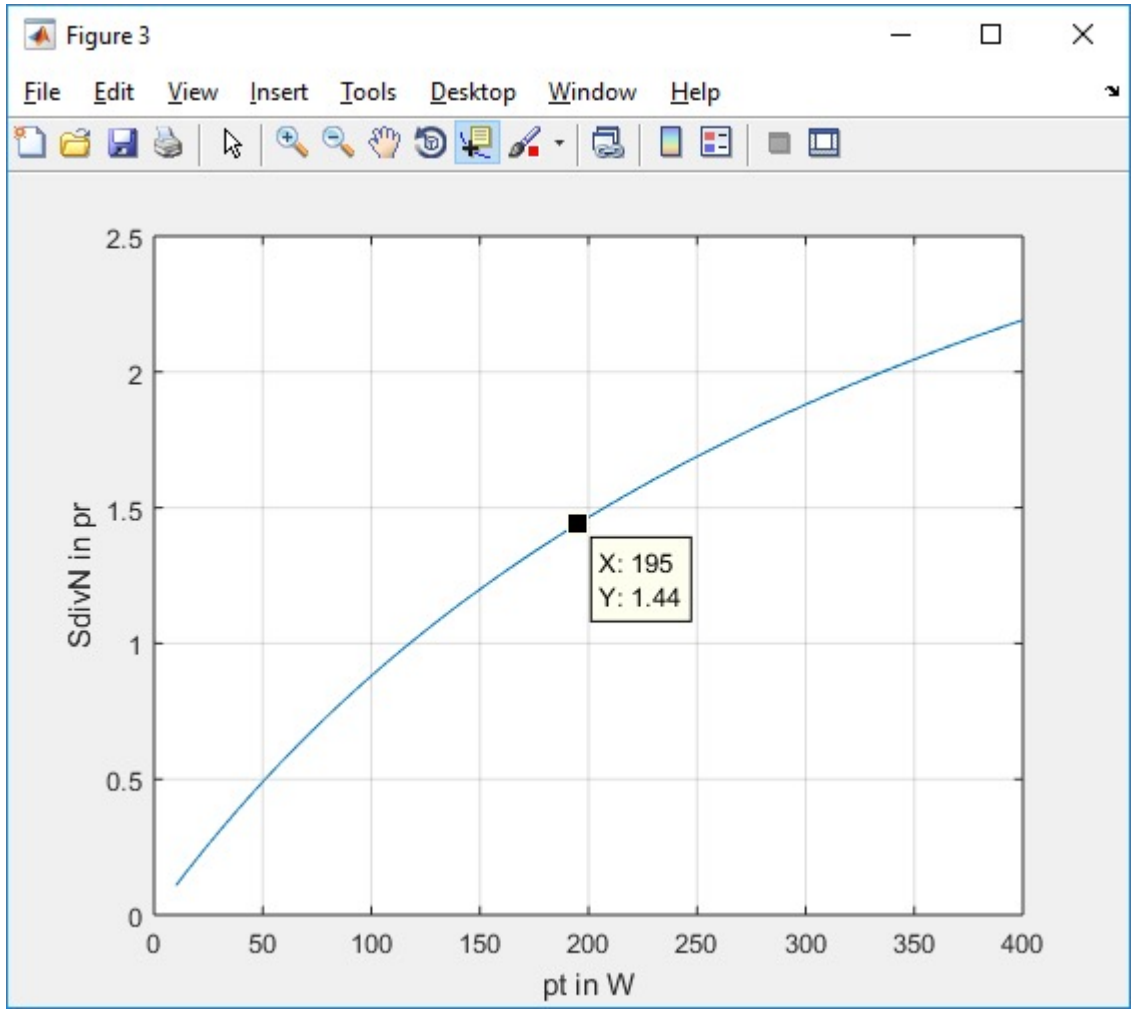
TRADE STUDY S/N vs Grazing Angle ψ



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see for ψ between 0 to 50 deg, S/N increases and Pd will increase as well. However; S/N depends on Pt as seen in the next slide. And the objective is not to have to much Pt Watts due to constrains in the satellite and also the higher the ψ the shorter Twarn

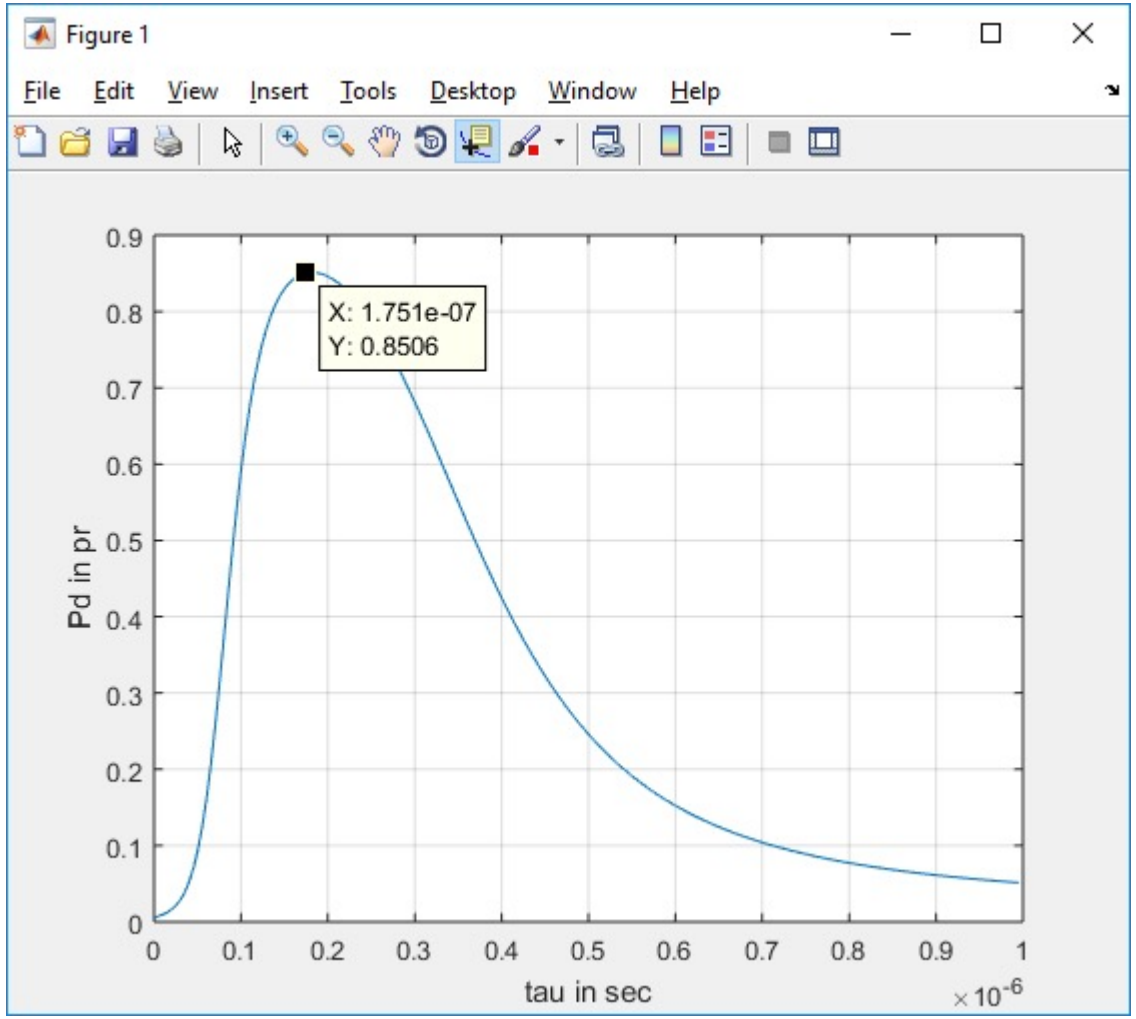
TRADE STUDY S/N or T/(C+N) vs Pt



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see the greater Pt is the greater T/(C+N)
We picked up Pt=195W for T/(C+N)=1.44 p.r.
The objective here is not to have too much Pt due to constrains in the satellite.

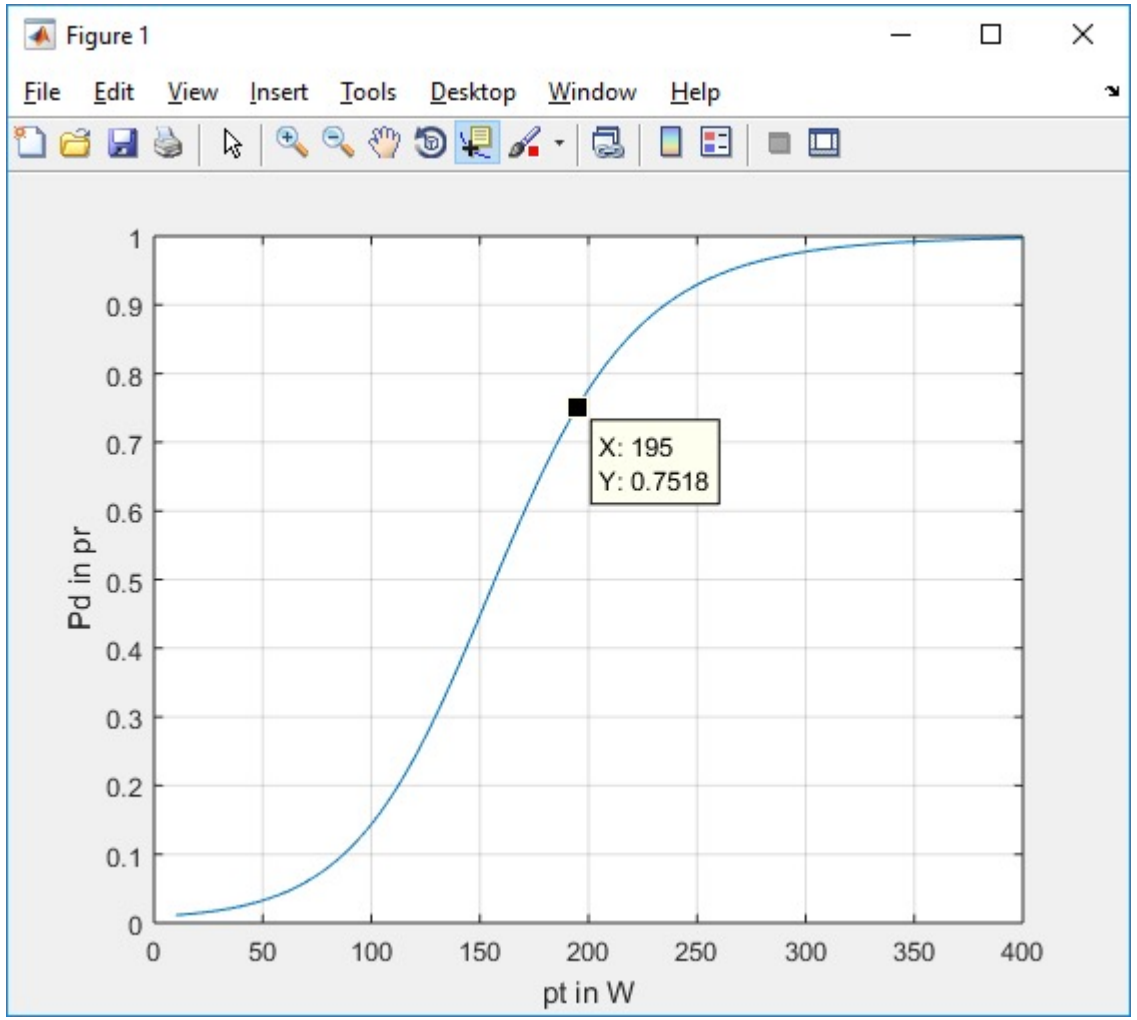
TRADE STUDY Pd vs tau



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see for tau=0.175microsec Pd is the highest. However due to constrain for dutycycle ≤ 1.0%
tau ≤ 0.1232 microsec
We picked up:
tau=0.1232 microsec
Then Pd=0.7518 p.r.

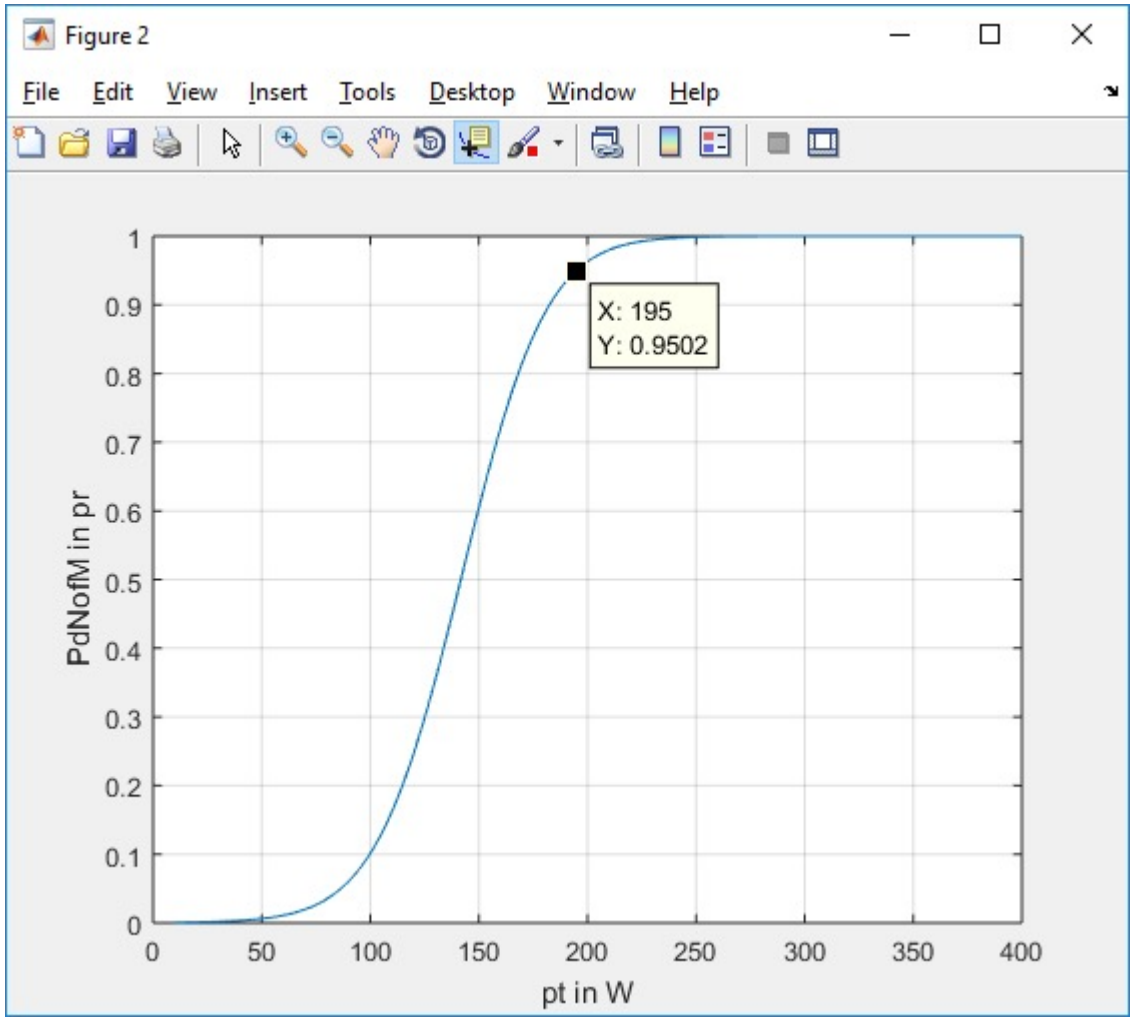
TRADE STUDY Pd vs Pt in Watts



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see the greater Pt is the greater Pd is for a single scan. (@boresight)
We picked up Pt=195W for Pd=0.7518 p.r.
The objective here is not to have too much Pt due to constrains in the satellite.

TRADE STUDY Pd Nof M vs Pt in Watts



For constant parameters:
Pt= XX W
PRF= XX pps
Bele=XX deg; Bazi=XX deg
Ant Spin Rate= XX RPM
Tfa= XX sec
tau= XX microsec
# scans n= XX

We can see the greater Pt is the greater Pd N of M is (N=2 M=4).
We picked up Pt=195W for Pd N of M =0.9502 p.r. (@ boresight).
The objective here is not to have too much Pt due to constrains in the satellite.

DOPPLER ANALYSIS

Relative Velocity

- This requirement must be verified by analysis using a worst-case scenario for the target speed and direction relative to the radar look direction. This analysis shall be presented in your proposal assuming the following worst-case conditions:
  - (1) max target speed and relative azimuth of 0° with respect to the radar look direction (approaching the radar);
  - (2) max target speed and relative azimuth of 180° with respect to the radar look direction (moving directly away from radar);
  - (3) max target speed and relative azimuth of 60° with respect to the radar look direction;
  - (4) 20% max target speed and relative azimuth of 0° with respect to the radar look direction;

Vrel(1)	71.9444	m/s
Vrel(2)	-8.0556	m/s
Vrel(3)	38.8904	m/s
Vrel(4)	39.9444	m/s

Doppler Frequency Target

$$v_r = v_{rel} * \cos(az) * \sin(cone\,angle)$$

v= vrel* cos(grazing)

$$f_d = 2 * v_r / \lambda$$

use this instead because Vrel has cos(az) already

Fd(1)	40563.40	Hz	Fd(1) near	40537.57	Hz	Fd(1) far	40589.10	Hz
Fd(2)	-4541.85		Fd(2) near	-4538.96		Fd(2) far	-4544.73	
Fd(3)	21927.00		Fd(3) near	21913.03		Fd(3) far	21940.89	
Fd(4)	22521.30		Fd(4) near	2.25E+04		Fd(4) far	22535.57	

SPEC COMPLIANCE MATRIX

List all Specs and Goals:

Note: This example may or may not correspond to this semester's spec

Item	Spec	Design	Margin
Grazing angle	≥ 2.0°	20.00	18.0
Az antenna beamwidth	≥ 0.2°	0.21	0.01
El antenna beamwidth	≥ 0.2°	0.47	0.27
Antenna spin rate	≤ 180 rpm	77.5	102.5
Target tracking	4 scans	8	4
Goal=	10 scans		- 2
Target detection capability			
Pdsingle scan	≥ 0.5	0.75	0.25
Pd(2 of 4) Goal (worst case)	≥0. 95	0.92	- 0.03
Warning time	≥ 60 sec	147.4	87.4
Goal=	≥ 300 sec		- 152.5

NOTE: POSITIVE MARGIN IS REQ'D

NOTE: NEG MARGIN IS OK FOR GOALS

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SPEC COMPLIANCE MATRIX CONT

List all Specs and Goals

Note: This examcorrespond to this semester's spec

Item	Spec	Design	Margin
Transmit power (peak max)	2000W	195	1805
Goal	≤ 10 W		- 185
Transmitter duty cycle	≤ 1.0 %	0.98	0.02
Pulse range resolution	≥ 10.0 m	19.6	9.6
Pulse azimuth resolution	≥ 10.0 m	153.1	143.1
<Tfa>	(derived)	7.7	
<Tfa> shall be ≥ 10 * 60/spin rate, sec	≥ 10.0	12.0	2.0
Target velocity meas(max)	≤ 40 m/s	37.1	2.9
Doppler measprecision	≤ 500 Hz	141.9	8.1
Goal	< 100 Hz		-41.92
Pulse Repetition Freq	≤ 300 kHz	81	219

NOTE: NEG MARGIN IS OK FOR GOALS

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